Manual

Material and Inventory

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Copyright

Contents

1 Material and Inventory

Possible fields of application

The material and inventory management function allows for material/batches to be managed as well as classified and grouped in material types. These material types specifically control data collection and processing in MES.

The function package also provides functions to edit and, if necessary, correct data of collected batches at a later point in time.

Implementation notes

The function package is used if you:

- would like to control system performance relating to batches subject to the material.
- need to correct data of recorded batches subsequently

Integration

The material type, which is a fundamental element of order data as well as of component data or the used batches, integrates the function package with a variety of function groups.

Functions

- Configuration of material types
  - Editing function of material type master data to control material-specific processing
- Configuration of assignment of materials to material types
  - Editing function to classify material in material types
- Manual or automatic assignment of batch numbers
  - Manual or automatic assignment of batches or batch numbers according to configuration
- Editing and correction of batches
  - Editing and correction functions for recorded batches

2 Material Types

Summary

Menu	Master data → Material → Material types
Transaction code	mtyp
Function authorization	mtyp

Usage

This function is used to create or modify material types in the system.

Integration

The system uses the material type and the settings stored with it to control the system behavior of objects to which the material type is assigned.

Selection criteria

Material type

Unique material type key

Type

Category of the material type

Material type name

Description or name of the material type (plain text)

Field description - general

Material type

Unique material type key

Lot size

Typical lot size in which batches of this material type are produced.

Material type name

Designation of the material type (plain text)

Type

Category of the material type

At the moment assigning a type serves the purpose of classification and logical combination of similar material types. Control of processing is not connected with this assignment.

Unit

Unit in which the materials of this type are listed.

Field descriptions – input batch processing

Inventory management

E - Yes, when logging input batch off
In this case the quantity of the batch consumed or the remaining quantity is to be collected manually when the batch is logged off. The quantity reduction and the transfer of the consumption posting to higher-level systems are performed only after the batch is logged off.

R - Yes, backflush (retrograde)

G - Yes, backflush (only for YIELD of the output batch)
like R; however, the quantity of the input material is only deducted in a retrograde manner if the output batch produced is a yield batch (batch class = G)

N - No inventory management. Batches of this material type are not considered in the stock overview.

Please note: Inputting the remaining quantity or the consumption depends on the configuration and is generally possible in an input batch change on the Windows terminal. In "Log input batch off" this quantity is posted to the respective batch. This applies for options E, R, G and N.

A – Anonymous inventory management
Batches of this material type are processed as material without batch reference in composition.

Logoff of input batches (when OP is interrupted/logged off)

A – Input batches are logged off automatically when an operation is interrupted or logged off and, finally, receive the batch status "processed".

F – Input batches are logged off automatically when an operation is interrupted or logged off and, finally, receive the batch status "free".

G – Input batches are logged off when an operation is interrupted or logged off as it is configured in the basic parameter settings.

N – Input batches are not logged off automatically when an operation is interrupted or logged off.

S – Input batches are logged off automatically when an operation is interrupted or logged off and, finally, receive the batch status "blocked".

Tolerances when logging off

Value in %.

When the input batch is logged off, a check is made as to whether or not the absolute value of the remaining quantity is less than the percentage value of the original quantity as specified here. If this is the case, the batch is set to batch status "Processed", so it is identified as consumed.

Maximum value

The maximum value limits the tolerance when logging off.

If the remaining quantity is less than the specified percentage value but greater than the maximum value, the batch is not set to batch status "Processed".

Options

Can be logged on several times to one machine

With this option, it is possible for an input batch to be logged on at multiple positions within the component list (BOM items) of an operation at a machine.

If the option is not set, if an input batch is already logged on at a position, another attempt to log it on to another position will be rejected with an error message.

Example:

The following material components are assigned to an operation:

BOM item **0001**	Material **487191**
BOM item 0002	Material 923948
BOM item **0003**	Material **487191**
BOM item 0004	Material 574857
BOM item **0005**	Material **487191**

If the option is set, batch 000007153 with material number 487191 can either be logged on to only one of the three positions 0001, 0003, 0005 or to two or even to all of these positions. If the batch is already logged on, during the second (and every other) logon of this batch a selection window with the positions of this material is displayed. For another logon, the operator must now select the position at which the batch is to be logged on. After confirmation, the batch is logged on to this (additional) position. However, if the operator selects a position at which the batch is already logged on, the labeling of the button is modified such that it is interpreted as a batch logoff.

In general, a multiple logon of different batches for the same materials (same material at different BOM items) is possible - regardless of how this option is set. If this option is set, the following option, *Parallel use allowed* is interpreted as being implicitly set.

Logging on unplanned material is not supported with this function.

Can be logged on to several machines at the same time

The input batch can be logged on to multiple workplaces and/ or operations in parallel at one time. Here it is does not make any difference if the input batch that is already logged on to an operation on a machine

- is logged on to another operation on the same machine
- is logged on to another operation on another machine
- is logged on to the same operation on another machine

Input batch must be logged on

This option activates an additional validation check during the logon of operations that require batch management. Here, for all of the material components (BOM items) which have a material type for which a check is activated, the system checks if there are already batches logged on. If this inspection fails, the logon of the operation is refused.

If all materials of the component list must always be logged on as input batches before the OP is allowed to be logged on, the option *Log OP on only if all components are logged on as inp. batches* in the *Output batch processing* tab can be used to achieve this result based on material types.

The validation check is made when logging on the OP and during an output batch change.

When an input batch is changed using the function *Input batch change*, no check is made.

That means that you can log off an input batch with required log on without registering a new one for that position. In order to carry out another output batch change, however, a valid batch must have been logged on for this position. Without a logged on batch, however, the operation can be logged off or interrupted.

Hand batch number down

If an input batch with a material type set with this option is logged on at order start, this batch number is used as the batch number for the output batch to be logged on.

At the same time, for the output batch logon a check is made as to whether or not the batch number of the input batch may be handed down, i.e. if the material type of the input batch is configured with this option.

The option is applied and configured in connection with throughput batch processing.

If this identifier is set, the identifier "Input batch only valid for exactly 1 output batch" must also be set.

Input batch only valid for exactly 1 output batch

For components with a material type for which this identifier is set, at least 1 batch per output batch must be logged on as an input batch.

Please note: The identifier only controls the processing during a batch change (CA_WL). It does not ensure that an input batch can be logged on again at a later time (at which another output batch is active).

Allow entry of unknown input batches

This option allows input batches previously unknown on the system side to be automatically generated and logged on.

If this option is enabled and an input batch that is unknown in the system is logged on, the material number of the component is automatically used for generation when logging the input batch on. The batch is created with a quantity of 1000000000 in the primary quantity unit of the OP and a goods receipt posting is performed.

In order to determine the right material type, there has to be a configuration/assignment between material and material type so that the material type can be determined correctly by way of the material number of the component. The *SYSTEM* material type is assigned automatically, provided that no assignment is kept for a material in HYDRA.

Use data for output material

This identifier specifies if the expiry date of the input batch can be used to calculate the expiry date of the output batch. Here it must be taken into consideration that only a logged on input material has this identifier based on its material type so that the output batch generated can accept the expiry date if it is configured accordingly (see "Determination of expiry date" in the "General" tab).

Decision on changing the input batch

If this option is enabled, a warning message will be displayed when changing the input batch before the running input batch is logged off.

Especially when serial numbers are collected, this warning message points out to the user that all individual components listed by serial numbers have to be evaluated before logging the input batch off in order for them to be "properly" connected to the resulting output batch (merged batch).

Pass batch attributes on

This option makes it possible to transfer configured batch attributes of an input batch to the resulting output batches. The configuration can be complemented flexibly by the advanced object configuration.

Pass document links on

This option makes it possible to transfer document links of an input batch to the resulting output batches. The configuration can be complemented flexibly by the advanced object configuration.

Field descriptions – output batch processing

Min. storage time

Min. storage time for the material type

During output batch determination, the corresponding availability date is calculated using this value (by adding it to the date/ time of manufacture). As long as this calculated time lies in the future, a batch remains in "Min. storage time" status and cannot be logged on.

Please note: Based on the material, this setting can be overridden using the configuration assignment of material to material type.

Warning limit

Warning limit of the material type

During batch generation, the corresponding warning date is calculated using this value (by adding it to the date/ time of manufacture). This can be used for an evaluation, e.g. using the function warning report.

Please note: Based on the material, this setting can be overridden using the configuration assignment of material to material type.

Expiry limit

Expiry limit of the material type

During batch generation, the corresponding expiry date is calculated using this value (by adding it to the date/ time of manufacture). At this point in time the batch is automatically set to "Expired" status, so it can no longer be registered.

Please note: Based on the material, this setting can be overridden using the configuration assignment of material to material type.

Reserved when batch is generated

This option can be used to automatically reserve the output batches produced for subsequent processes. The reservation can be made on different levels.

Values:

- - No automatic reservation for batch generation
- - Reservation for all OPs of the order
- - Reservation for subsequent OPs

With the last two values, for a subsequent input batch logon a corresponding validation check for the existing reservation is made.
In reel-based production (MPL-RF) on so-called cutting machines the reservation is always specified in relation to the subsequent parent operation.

Options

Enter input quantity in relation to batches (MPLRF-BP)

Please note: This option should only be set as an exception, e.g. due to requirements resulting from the project.

If this option is set, the input quantity is collected in relation to the output batch. If this option is not set, the input quantity is collected in relation to the operation.

Input batch change obligatory for residual quantity <= 0

Provided that this option is set, those input batches that are currently logged on are highlighted in red the residual quantity of which is less than or equal 0, when output batches are changed. These input batches now have to be logged off.

Please note: This option is only taken into account for MPL machines (batch management) assigned to the machine type "M". Processing takes places locally at the terminal.

Log OP on only if all components are logged on as inp. batches

If this option is set, the OP can only be logged on for the output material when all of the materials of the component list are logged on as input batches.

If this option is not set, the OP can still be logged on if not all of the materials are logged on as input batches (yet).

Consumption balance

During the OP logoff, an additional dialog (V_BLZ) is opened displaying the material components and their consumption quantities in relation to the current OP logon. In this dialog, the operator has the option to log off input batches that are still running. The option only becomes active when the consumption balance on the machine was activated in the MPL tab.

Serial no. obligation (MPL-SNR, starting with MPL 7.2)

In the collection of serial numbers (ADE-SNR), when this option is active and the operations require batch management, an additional batch per serial number is created in the MPL module. The collection of the batches per serial number (using command A_TR) is performed only for operations with serial number obligation = E. In the batch relationships (mpl_beziehungen) table, traceability to the current output batch (ID) is established per collected serial number.

Generation of HU (MPL-SNR, starting with MPL 7.2)

If the serial number obligation option is set, this option can be used for additional control with respect to whether or not the collected serial numbers are stored in stock in relation to a handling unit (merged batch). The option is currently only intended for customizations and does not have any effect. BUL: During an output batch change, the acronyms CNRHU:G / CNRHU:A / CNRGENHU:G / CNRGENHU:A can be used to generate a HU for serial numbers as a yield or scrap HU. In this case, the serial numbers are assigned to the HU batches (merged batches) and in the relationship table (mpl_beziehungen) the traceability to the current output batch (ID) is created.

Please note that this option is only checked at OP logon and output batch change. When an input batch is changed using the function *Input batch change*, no check is made.

Log off/interrupt OP without the last batch

If the option is set, a last output batch will indeed be generated for the article of the operation (matching this material type) between the last output batch change and/or OP logon and OP interruption and/or OP logoff, but this one will be deleted immediately (batch status = "D"). The batch is not visible in the batch history.

Delete batch assignment

If this flag is set, the connection to the running input batches of the OP will be deleted for the output batch "deleted at last". Consequently, the deleted batch is neither visible within batch tracing.

Automatic assignment of serial numbers

This option enables the automatic assignment of numbers for a new part when merging components listed by serial numbers. If this option is not set, the serial number may be assigned manually. The option is only relevant if the flag "Superordinate serial number" = N is set for the component.

User field key for output batches

The defined user field key is transferred to the generated output batches.

Field description - general

Determination of expiry date

- **Current point in time** – the expiry date is calculated based on the current point in time and the duration stored with the material type of the output batch.
- **Shift date** - The expiry date of the generated output batch is determined based on the current shift date, the time 00:00 and the expiry limit stored with the material type.
- **Input batch** The expiry date of the generated output batch is determined from the running input batch where the option "Use data for output material" is set on the material type.
- **Processing according to the advanced object configuration** The determination of the expiry date of the generated output batch is specified using an additional configuration in the advanced object configuration:

- **<u>Use of the closest expiry date of all logged on input batches</u> -** If this configuration is active, the expiry and warning time of the input batch with the closest expiry time is accepted in the output batch. The availability date is calculated using the configuration in the "Output batch processing" tab.

Parameter name	Parameter value
Object type	MATTYP
Object ID 1	Material type of the output batch for which the configuration applies
Parameter	DATE_OF_EXPIRY_CALCULATION
Parameter value	MIN_DOE_OF_INPUT_MATERIAL

- **Use of the closest expiry date of selected input batches** If this configuration is active, the expiry and warning time of the input batch with the closest expiry time is accepted in the output batch. The availability date is calculated using the configuration in the "Output batch processing" tab.

Parameter name	Parameter value
Object type	MATTYP
Object ID 1	Material type of the output batch for which the configuration applies
Parameter	DATE_OF_EXPIRY_CALCULATION
Parameter value	MIN_DOE_OF_SELECTED_INPUT_MATERIAL
In the determination of the relevant input batches, however, only those batches are considered that are identified using an additional configuration related to the material type.	
Parameter name	**Parameter value**
Object type	MATTYP
Object ID 1	Material type for which the configuration applies
Parameter	DATE_OF_EXPIRY_CALCULATION
Parameter value	INCLUDE_IN_DOE_SELECTION

WIP material (Work in Process)

This flag defines whether a batch of this material type is to be logged on automatically as input batch for the subsequent operation (on the basis of the operation sequence within a production order) as WIP material. The configuration for the function is described here.

Options

Transfer to interface (goods movements)

If this indicator is set, the produced output batch is identified as goods receipt or the used up input batch as goods issue and as being subject to uploads within the material movements.

Hold back goods receipt until usage decision is made

You can use this option to delay the transfer of goods receipts to the PPS system. In this case, the output batch is transferred only if a usage decision is taken.

Field descriptions - transport

Generate transport order for output material

This option creates a transport order relating to batches for a generated output batch. The transport is started from the material buffer in which the output batch is produced. The option set here overrides the configuration of the relevant option within the configuration of workplaces.

Generate transport order for input material

This option creates an article-related transport order in relation to a material component, when an operation is planned for a machine using the shop floor scheduling module. Transportation is started from the output material buffer of the preceding operation. The option set here overrides the configuration of the relevant option within the configuration of workplaces.

Field descriptions – user fields

Customizing can be used to release user fields for the object type “MATTYP”.

3 1 Assignment Material - Material Type

Overview

Menu	Master data → Material → Material assignment - Material type
Transaction code	asmm
Function authorization	asmm

Usage

This function is used to assign the appropriate material type to a material in the system.

Integration

Each material can be assigned to a material type. The assignment is used

- if a goods receipt batch is recorded manually
- if an unknown batch is defined

Requirement

The material type must be defined in the system.

Selection criteria

The following selection criteria are available in the application:

Material

Only the selected materials are selected.

Material type

Only materials with the selected material type are selected.

When using multiple selection criteria - if nothing else is specified - the amount of overlap of the selection criteria is displayed.

Field descriptions

Material

Material number that is to be assigned to the material type.

Material type

Assigned material type. This must be defined in the system.

Comment

Additional text regarding the material, comments

Min. storage time

Minimum storage time for the material type

During batch determination, the corresponding availability date is calculated using this value (by adding it to the date of manufacture). Up to that point, a batch remains in "Min. storage time" status and cannot be registered.

Please note: this setting overrides the setting with the same name in the *Material type* configuration.

Warning limit

Warning limit of the material type

During batch determination, the corresponding warning date is calculated using this value (by adding it to the date of manufacture). This can be used for an evaluation, e.g. using the function *Warning report*.

Please note: this setting overrides the setting with the same name in the *Material type* configuration.

Expiry limit

Expiry limit of the material type

During batch determination, the corresponding expiry date is calculated using this value (by adding it to the date of manufacture). At this point in time the batch is automatically set to "Expired" status, so it can no longer be registered.

this setting overrides the setting with the same name in the *Material type* configuration.

Notes

A material can only be assigned to exactly one material type. However, one material type can be assigned several materials. The saving procedure for material is case sensitive. The so-called wildcards "*" and "?" can be used in the material field. The selected material type must exist during creation. Otherwise, the creation will be refused.

If a material is assigned to another material type, the material type is NOT automatically updated
- for existing orders/operations
- for existing component lists
- in the existing batches

4 Generating Batch Numbers

Usage

The batch number is generated to basis 35; the digit specifications are relative and refer to the fixed proportion. The fixed proportion can be deduced from the prefixes for the automatically generated batch numbers defined in the basic settings.

Digit 1-2	Terminal number
Digit 3	last digit of the year (starting at 1998)
Digit 4-5	Day of the year
Digit 6...	running number, distributed along the length of the batch number

The number generated at the terminals is unique for each terminal, i.e. there is no batch number 4711 for a goods receipt batch (create a batch) and a production batch (output batch change) on one day in a year.

5 Batch Data Overview

Summary

Menu	Material management → Inventory management → Batch data overview
Transaction code	batov
Function authorization	batov

Usage

The batch data overview is used to display or process one or more batches depending on the chosen selection criteria entered.

Integration

The batch data overview shows all batches included in the data set that were specified by the selection criteria entered.

In the batch data overview, the selected batches can be displayed in their current status together with detail information.

Selection criteria

The application provides the following selection criteria:

"Batch" category

Batch number

The batch number represents the batch ID used by the user. This can be:

- for the input/output at terminal dialogs
- for tracing or information search at the office client

When the configuration option "automatic generation of batch no. when creating batches" is activated, a batch number will still only be assigned automatically as long as the batch number field in the editing dialog box is left empty when a batch is added.

Material

Entering the material number as a selection criterion displays all batches that have this material number.

Workplace

Entering the workplace as a selection criterion displays all batches that have been produced at this workplace/machine.

MES order number

MES order number (order/operation number) by which the batch has been produced.

Internal batch number

The batch number is a unique, system-wide batch identification number.

Material buffer

The individual batches are located in a specific material buffer. With this selection, all batches with the selected material buffer are displayed.

Material type

The individual batches or materials belong to a material type. The same transport and handling guidelines are used for these material types across the system. With this selection, all batches with the selected material type are displayed.

Material category (type)

The material type is used to assign batches to specific classes/ groups. With this selection, all batches with the selected material type are displayed.

Material designation

When the output batch is created on the shop floor stations, the material designation of the currently registered order is used. Because no material master is managed, the material designation is saved redundantly in the batch description.

Manufacturing date from / until

Date/period of the production of a batch

Consider long-term data

Archived data may be selected.

"Status" category

Batch class

The batch class describes the overall quality of the batch. With this selection, all batches matching the selected batch class are displayed.

Batch status

The batch status describes the technical system and production status of a batch. Selecting the batch status as a selection criterion displays all batches that have this status.

Quality status

The quality status "blocked" prevents a batch from being logged on. Selecting the quality status as a selection criterion displays all batches that have this status.

Manual Q status

Selecting the manual Q status as a selection criterion displays all batches that have this status.

Material status

The material status indicates a logical status of the batch, e.g. packed, tested. Selecting the material status as a selection criterion displays all batches that have this status.

Transport status

The transport status represents the technical system status with respect to transfer postings to external storage. Selecting the transport status as a selection criterion displays all batches that have this status.

Advance logon flag

Restriction to batches logged on in advance.

"Attributes" category

Attribute (1 to 10)

The data displayed may be restricted to the batch attributes directly kept for the batch.

"Batch attributes" category

Batch attribute (designation)

The data displayed may be restricted to the batch attributes configured for the material type. There are 40 text fields, 20 numeric fields and 20 decimal fields that may be configured altogether.

"Alternative batch numbers" category

Alternative batch number (1 to 20)

Selecting an alternative batch number as a selection criterion displays all batches that currently have this identifier.

"Reservation" category

Reserved for order

Entering or selecting the order number in the field displays all batches that were produced for this order/ OP.

Reserved for OP

Entering or selecting the order/OP number in the field displays all batches that were produced for this order/ OP.

"Dates" category

Expiry date from / until

Date/period that indicates the shelf life of a batch

Availability date from / until

Date/period of the availability of a batch

Warning time

Warning date of a batch

"Miscellaneous" category

Serial number

Selecting a serial number as a selection criterion displays all batches for which this serial number is entered in the serial number field.

Batch (LOT) number

Selecting the batch/lot number as a selection criterion displays all batches that currently have this identifier.

Person

Selecting the person as a selection criterion displays all batches that have been produced by this person and, as a result, are currently assigned to this person (identifier).

Merged batch

Selecting the merged batch number as a selection criterion displays all batches for which the merged batch number is entered in the merged batch field.

PPS batch

Selecting the PPS batch as a selection criterion displays all batches for which the PPS batch number is entered in the PPS batch field.

Editing functions

These functions are provided by the standard features for creating, editing, etc. to edit one or several data records:

 **Add batch**

This function can be used to insert a new batch. Goods movements are not generated for the new batch.

 **Copy batch**

This function can be used to copy a batch. However, the batch number for the new batch has to be entered anew. Goods movements are not generated for the new batch.

 **Edit batch**

A batch can be edited using this function. But neither cancellations nor goods movements are generated for the changed batch.

 **Delete batch**

This function can be used to delete a batch. But neither cancellations nor goods movements are generated for the deleted batch.

Pool batches (new batch number)

Pooling/merging batches. A new batch number is generated for the new batch (merging batches).

When pooling merged batches, all individual batches are assigned to a new merged batch number.

Pool batches (use existing batch numbers)

Pooling/merging batches. A batch number already included in the pool/merge is used for the new batch (merging batches).

When pooling merged batches, all individual batches are assigned to the selected merged batch number that is part of the pool.

Split batch

A new batch is created by splitting it off from an existing batch (splitting batches). In this case, the following options exist for the remaining target quantity of the original batch:

- Repost remaining quantity of batch to new batch
- Reduce existing batch to the remaining quantity

If a merged batch is split, the selected individual batches/serial numbers are transferred to a split off batch. The remaining individual batches/serial numbers remain assigned to the original merged batch.

Repost

This function can be used to repost a batch to another material buffer (Reposting batches).

Generate

The function "enter goods receipt batch" is used to create batches manually (Generating batches). This is necessary if material is delivered via the goods receipt, for example.

Edit batch attribute

The batch attributes of the batch selected in the grid are shown in the tab “batch attributes”. Only those batch attributes will be shown that have been assigned the indicator “show attribute on client” within the configuration of batch attributes.

Provided that batch attributes are shown, they can be edited by clicking the button “edit batch attribute”. The field types/field lengths specified for the configuration of batch attributes are not checked.

Go to

Graphic batch tracing

Starts the application Graphic batch tracing.

Document management

Starts the document management

"Batch data overview" detail application

In the batch data overview detail application, all batches are displayed according to the selections entered.

The information or field descriptions of the selected batches and, thus, of the detail application are described in the documents entitled batch object and batch structure.